Quantitative Portfolio Optimization & Risk Analysis

Modern Portfolio Theory (MPT) Application Across an 8-Asset Universe (2019–2024)

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Executive Overview & Project Purpose

<u>Core Objective:</u>

- Built an institutional asset allocation model to evaluate how mathematical optimization balances risk and return across market cycles (2019–2024).
- Evaluated an 8-asset universe spanning US Equities (SPY), International (EFA), Emerging Markets (EEM), Aggregate Bonds (AGG), Long Treasuries (TLT), Real Estate (VNQ), Gold (GLD), and Small-Caps (IWM).

<u>Role of the Excel Model:</u>

- Serves as the quantitative engine behind these findings, housing the raw time-series data, log return calculations, variance-covariance matrices, and automated Solver constraints.

The Quantitative Engine Architecture

Translating Historical Market Data into Algorithmic Allocation Strategies:

- Data Pipeline: Standardized 60 monthly price observations (2019–2024) into continuous log returns (=LN) to establish statistical baselines.
- Covariance Engine: Leveraged Excel array algebra (MMULT, TRANSPOSE) alongside a 8 X 8 Variance-Covariance matrix to account for cross-asset correlations.
- Non-Linear Optimization: Configured Excel Solver (GRG Nonlinear) to isolate optimal portfolio weights for Conservative, Moderate, and Aggressive risk mandates.
- Frontier Mapping: Programmatically derived the Markowitz Efficient Frontier curve to identify the Global Minimum Variance and Tangency (Max Sharpe) portfolios.

Performance Outcomes & Stress Test Results

Maximizing Risk-Adjusted Returns Through Cross-Asset Diversification:

- Sharpe Ratio Expansion: The optimized Moderate (Max Sharpe) Portfolio achieved a 10.2% expected return with 13.4% volatility, yielding a 0.46 Sharpe Ratio (vs. 0.27 for an equal-weighted baseline)
- Primary Risk Buffers: Heavy allocations to SPY (57.6%) and Physical Gold (GLD, 42.4%) drove variance reduction due to low historical co-movement between equities and bullion.
- Downside Cushioning (2020 COVID Crash): While the SPY benchmark dropped -19.6%, the optimized Moderate Portfolio limited drawdowns to -6.2% (+13.4% outperformance).
- Rate Shock Resilience (2022 Inflation): While SPY fell -18.1%, the Moderate Portfolio held drawdown to -8.9% (+9.2% outperformance) by avoiding long-duration Treasuries (TLT).